

## Demonstration of Proposed Features

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 15 July 2026                          |
| <b>Team ID</b>       | XXXXXX                                |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 1 Mark                                |

### Demonstration of Proposed Features:

| S.No | Feature Name                 | Description                                                                                                                                                           | Status<br>(Implemented /<br>Partial / Pending) | Demonstrated<br>(Yes / No) | Remarks                                         |
|------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------|-------------------------------------------------|
| 1    | User Input Interface         | Accepts socio-economic indicators such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita through a user-friendly web form. | Implemented                                    | Yes                        | Successfully accepts user inputs.               |
| 2    | Data Preprocessing           | Cleans and preprocesses the dataset before training by handling missing values and preparing features.                                                                | Implemented                                    | Yes                        | Improved model quality and prediction accuracy. |
| 3    | Machine Learning Prediction  | Predicts the Human Development Index using the trained Machine Learning model.                                                                                        | Implemented                                    | Yes                        | Prediction model integrated successfully.       |
| 4    | Prediction Result Display    | Displays the predicted HDI value and development category instantly on the results page.                                                                              | Implemented                                    | Yes                        | Real-time prediction displayed successfully.    |
| 5    | Model Performance Evaluation | Evaluates the trained model using appropriate regression evaluation metrics to ensure reliable predictions.                                                           | Implemented                                    | Yes                        | Model performance verified successfully.        |

|   |                       |                                                                                                                     |             |     |                                                         |
|---|-----------------------|---------------------------------------------------------------------------------------------------------------------|-------------|-----|---------------------------------------------------------|
| 6 | Flask Web Application | Integrates the trained HDI model into a Flask web application for real-time prediction through a browser interface. | Implemented | Yes | Complete end-to-end workflow demonstrated successfully. |
|---|-----------------------|---------------------------------------------------------------------------------------------------------------------|-------------|-----|---------------------------------------------------------|

**Feature Implementation Summary:**

| Feature                     | Value |
|-----------------------------|-------|
| Total Features Proposed     | 6     |
| Total Features Implemented  | 6     |
| Total Features Demonstrated | 6     |